MDM Project Report on

AUTOMATED ON/OFF SWITCH FOR WATERLINE CONTROL

(Domain- Home Automation)

Submitted in partial fulfillment of the requirements

*of the **Third Year** of*

Bachelor of Technology

by

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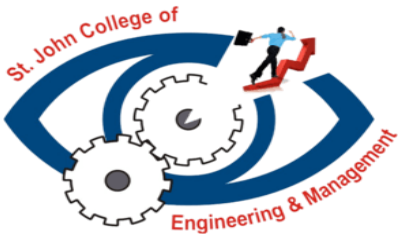
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ABSTRACT

Waterline operation in residential buildings often depends on manual switching of pumps and valves, requiring physical presence and continuous attention from users. This can lead to inconvenience, unnecessary water flow, water wastage, and difficulty in controlling the water supply remotely. Traditional waterline systems generally rely on manually operated switches or conventional valve mechanisms and provide limited remote control capabilities. To address these problems, the proposed project, **"AUTOMATED ON/OFF SWITCH FOR WATERLINE CONTROL,"** provides an IoT-enabled solution for remotely controlling a domestic waterline through a mobile application.

The objective of the proposed system is to simplify waterline operation by integrating an **ESP32 DevKit module, Wi-Fi communication, an L298N motor driver, a 12V DC geared motor rated at 30 RPM, and a motor-operated ball valve**. The system allows the user to send ON/OFF commands through a mobile application. The command is transmitted through a Wi-Fi network to the ESP32 controller, which processes the command and controls the motor driver. The 12V geared motor then rotates the connected ball valve mechanism to open or close the waterline. This eliminates the need for direct manual operation and enables convenient remote control from a mobile device. The proposed system combines concepts of **IoT, home automation, embedded systems, electrical control, and mechanical actuation**. The ESP32 DevKit acts as the main controller and provides Wi-Fi-based communication between the mobile application and the hardware. The L298N motor driver controls the direction of motor rotation, while the 12V, 30 RPM geared motor provides the mechanical movement required to operate the ball valve. A **12V DC adapter** supplies power to the system, and a **step-down converter** reduces the voltage to 5V for powering the ESP32 DevKit. The mechanical arrangement converts the motor's rotational movement into the required valve ON/OFF operation. The system can also be extended with water-level, flow, or other sensors for monitoring and intelligent control.

The project follows the **Design Thinking methodology**, beginning with the **Empathize** stage to understand problems associated with manual waterline operation, followed by **Define** to identify the engineering problem and user requirements. During the **Ideate** stage, different control approaches such as manual automation, timer-based control, water-level-based control, and IoT-based control were considered. The selected concept was then developed into a **Prototype** integrating the mobile application, ESP32 DevKit, L298N motor driver, 12V geared motor, ball valve, step-down converter, and 12V power adapter. The **Test** stage focuses on validating remote ON/OFF operation, Wi-Fi communication reliability, motor response, valve movement, and safe electrical and mechanical operation. The overall system architecture integrates mobile control, IoT communication, motor driving, power conversion, and mechanical valve actuation.

Thus, the **Automated ON/OFF Switch for Waterline Control** is a practical **IoT-based home automation solution** that converts a conventional manually operated waterline into a remotely controllable system. The project demonstrates the practical application of ESP32-based IoT control, L298N motor driving, 12V geared motor integration, regulated 5V power supply, and motorized ball-valve operation. By enabling users to control the waterline through a mobile application, the proposed system aims to reduce manual effort, improve convenience, and support more efficient waterline operation.

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ABBREVIATIONS AND SYMBOLS

- **IoT:** Internet of Things
- **ESP32:** Espressif 32-bit Microcontroller
- **Wi-Fi:** Wireless Fidelity
- **MCU:** Microcontroller Unit
- **API:** Application Programming Interface
- **UI:** User Interface
- **UX:** User Experience
- **DFD:** Data Flow Diagram
- **UML:** Unified Modeling Language
- **AC:** Alternating Current
- **DC:** Direct Current
- **RPM:** Revolutions Per Minute
- **V:** Voltage
- **A:** Ampere
- **W:** Watt
- **HP:** Horsepower
- **LED:** Light Emitting Diode
- **NO:** Normally Open
- **NC:** Normally Closed
- **COM:** Common Terminal
- **GND:** Ground
- **GPIO:** General Purpose Input/Output
- **PWM:** Pulse Width Modulation
- **HTTP:** HyperText Transfer Protocol
- **IP:** Internet Protocol
- **LAN:** Local Area Network
- **MCB:** Miniature Circuit Breaker
- **PCB:** Printed Circuit Board
- **AC:** Alternating Current
- **ON/OFF:** Open/Close Control State
- **IoT:** Internet of Things
- **Q:** Water Flow Rate
- **A:** Cross-sectional Area
- **V:** Flow Velocity

CHAPTER 1: OVERVIEW

1.1 Introduction

Water is an essential resource in residential buildings, and its proper distribution and control are important for daily household activities. In conventional waterline systems, the operation of water valves and pumps is generally performed manually. The user needs to be physically present near the switch or valve to start or stop the water supply. This manual operation can cause inconvenience, unnecessary water flow, and difficulty in managing the waterline when the user is away from home.

With the development of **Internet of Things (IoT), embedded systems, and home automation technologies**, conventional household operations can be transformed into smart and remotely controllable systems. IoT enables electronic devices to communicate through a network and allows users to control devices remotely using mobile applications.

The proposed project, **"AUTOMATED ON/OFF SWITCH FOR WATERLINE CONTROL,"** focuses on developing an IoT-based system for remotely controlling a domestic waterline. The system uses an **ESP32 DevKit module with built-in Wi-Fi, an L298N motor driver, a 12V DC geared motor rated at 30 RPM, and a motor-operated ball valve**. The user can send an ON or OFF command through a mobile application. The command is transmitted through Wi-Fi to the ESP32, which controls the motor driver. The motor rotates the connected ball valve mechanism to open or close the waterline, thereby providing automated water-flow control.

The project combines **Mechanical Engineering, IoT, Embedded Systems, Electrical Control, and Home Automation** concepts. The mechanical mechanism converts motor rotation into valve movement, while the ESP32 DevKit and L298N motor driver provide electronic control. A **12V DC adapter** supplies power to the system, and a **step-down converter** provides a regulated 5V supply for the ESP32 DevKit. The system is designed to reduce manual effort and provide convenient remote operation of the waterline.

The project follows the **Design Thinking approach**, beginning with understanding the user's problems and requirements through the **Empathize** stage, defining the engineering problem, generating and selecting suitable concepts during **Ideate**, developing the selected concept as a **Prototype**, and validating its operation through **Testing**. The final solution integrates mobile-based IoT control, ESP32-based communication, motor-driver control, power conversion, and mechanical ball-valve actuation to provide a practical and user-friendly automated waterline control system.

1.2 Background

In conventional residential waterline systems, water flow is commonly controlled using manually operated switches and valves. The user is required to be physically present to operate the system, which can become inconvenient when the water supply needs to be controlled from a different location. Manual operation may also result in the waterline remaining open for longer than required, leading to unnecessary water flow and potential water wastage.

The increasing use of Home Automation and Internet of Things (IoT) technologies provides an opportunity to improve conventional waterline control systems. IoT-based systems allow devices to communicate through a network and provide remote control through a mobile application. In the proposed system, an **ESP32 DevKit module** is used as the central IoT control unit. Its built-in Wi-Fi capability enables communication between the mobile application and the hardware.

The proposed system integrates an **L298N motor driver and a motorized ball-valve mechanism**. The L298N motor driver controls the direction of rotation of the motor, while the

1.3 Importance of the Project

The project is important as it provides **remote and convenient waterline control** using IoT technology. It reduces manual effort and helps minimize unnecessary water flow through timely ON/OFF operation.

Key benefits:

- Remote control through mobile application.
- Reduces manual operation.
- Uses ESP32 and Wi-Fi for IoT control.
- Provides motorized valve operation.
- Helps reduce water wastage.
- Low-cost and practical home automation solution.
- Can be extended with sensors and smart monitoring.

1.4 Perspective of Stakeholders and Customers

The **Automated ON/OFF Switch for Waterline Control** provides benefits to different stakeholders involved in domestic water management.

Homeowners / Users

Users can remotely switch the waterline **ON/OFF through a mobile application**, reducing manual effort and providing convenient control.

Family Members

Family members can operate the system remotely when required, reducing the need to be physically present near the waterline.

Building Caretakers

Caretakers can easily control waterline operation and reduce unnecessary running of the system, especially in residential buildings.

Plumbers / Maintenance Personnel

The system provides a simpler automated control arrangement, making operation and maintenance easier.

Society / Building Management

The system can help improve water management by reducing unnecessary water flow and providing a practical **IoT-based home automation solution**.

Customers

Customers benefit from **remote operation, convenience, reduced manual intervention, and scope for future smart monitoring**.

1.5 Objectives and Scope of the Project

Objectives

1. To provide **remote ON/OFF control** of the waterline through a mobile application.
2. To integrate **ESP32, Wi-Fi and motorized valve** for automated operation.
3. To reduce **manual effort and unnecessary water flow** through convenient control.
4. To develop a **low-cost IoT-based home automation solution**.
5. To provide a system that can be extended with **sensors and smart monitoring** in the future.

Scope

The project focuses on **domestic waterline automation** using a mobile application, ESP32, motor, and valve mechanism. The system enables remote ON/OFF operation and can be further extended with **water-level/flow sensors, alerts, and intelligent automation** for improved water management.

1.6 Summary

This chapter provided an overview of the proposed **"Automated ON/OFF Switch for Waterline Control"** project. It discussed the problems associated with conventional manual waterline operation, such as the need for physical presence, inconvenience, and unnecessary water flow. The proposed system addresses these problems by integrating **IoT and home automation technologies**.

The system uses an **ESP32 DevKit module, Wi-Fi communication, an L298N motor driver, a 12V DC geared motor rated at 30 RPM, and a motorized ball valve** to provide remote ON/OFF control through a mobile application. A 12V DC adapter supplies power to the system, while a step-down converter provides a regulated 5V supply for the ESP32 DevKit.

The importance of the project, stakeholder requirements, objectives, and scope were also discussed. The system aims to reduce manual effort and provide a practical and low-cost approach to automated waterline control through remote operation and motorized valve actuation.

The project follows the **Design Thinking approach**, including **Empathize, Define, Ideate, Prototype, and Test**, leading to the development of the final IoT-based waterline control solution. The next chapter presents the **Literature Survey, Problem Definition, Feasibility Study, and Design Thinking Methodology** used for the development of the proposed system.



figure 1

CHAPTER 2: SURVEY & PROPOSED WORK

2.1 Introduction

The development of an **IoT-based Automated ON/OFF Switch for Waterline Control** requires an understanding of conventional waterline operation, home automation, IoT-based control, and motorized valve mechanisms. Traditional residential waterline systems mainly depend on manually operated switches and valves, which require physical presence and provide limited remote control capabilities.

The proposed project aims to overcome these limitations by integrating an **ESP32 DevKit module, Wi-Fi communication, an L298N motor driver, a 12V DC geared motor rated at 30 RPM, a plastic ball valve, and a mobile application**. The ESP32 receives commands from the mobile application through Wi-Fi and sends control signals to the L298N motor driver. The motor driver controls the direction of motor rotation, enabling the geared motor to operate the connected plastic ball valve between its open and closed positions. A **12V DC adapter** supplies power to the system, while a **step-down converter** provides a regulated 5V supply for the ESP32 DevKit.

This chapter presents the **problem definition, feasibility study, and Design Thinking methodology** used for developing the proposed system. The Design Thinking process includes **Empathize, Define, Ideate, Prototype, and Test**, helping to understand user requirements, identify the problem, select a suitable solution, develop the prototype, and evaluate its operation. The proposed system architecture integrates the **Mobile Application, Wi-Fi Network, ESP32 DevKit, L298N Motor Driver, 12V Geared Motor, and Plastic Ball Valve** to provide a practical and convenient IoT-based solution for remote waterline control.

2.2 DESIGN THINKING METHODOLOGY TABLE

Sr. No.	Design Thinking Stage	Activities Performed	Outcome
1	Empathize	Interviews were conducted with users to understand difficulties in conventional waterline operation, manual valve handling, and the need for remote control.	User problems and requirements were identified.
2	Define	The information collected through interviews was analysed to define the main problem of inconvenient manual waterline operation.	A clear problem statement was formulated.
3	Ideate	Different solutions were considered, including manual control, timer-based automation, sensor-based control, and IoT-based mobile control.	IoT-based remote waterline control was selected as the suitable concept.
4	Prototype	A prototype was developed using the ESP32 DevKit, L298N motor driver, 12V DC geared motor (30 RPM), plastic ball valve, step-down converter, and 12V adapter.	A hardware prototype integrating electronic control and mechanical valve operation was developed.

5	Test	The prototype was evaluated for mobile ON/OFF commands, Wi-Fi communication, motor rotation, and ball-valve operation.	The system's operation and areas requiring improvement were identified.

table 1

2.3 Problem Definition

Conventional domestic waterline systems generally require **manual operation of switches and valves**, requiring the user to be physically present to control the water supply. This can cause inconvenience when waterline operation is required from a different location and may lead to unnecessary water flow.

The problem is to design and develop a low-cost, user-friendly, and IoT-based automated waterline control system that enables users to remotely operate a plastic ball valve through a mobile application. The problem was considered from the user's perspective as part of the Design Thinking and MDMD Project approach, focusing on convenience, usability, affordability, and practical implementation.

The proposed system integrates the following components:

- **Mobile Application:** Provides ON/OFF commands for remote operation.
- **ESP32 DevKit Module:** Processes commands and communicates through built-in Wi-Fi.
- **L298N Motor Driver:** Controls the direction of rotation of the DC geared motor.
- **12V DC Geared Motor (30 RPM):** Provides mechanical rotation to operate the valve mechanism.
- **Plastic Ball Valve:** Controls the opening and closing of the waterline.
- **12V DC Adapter:** Supplies power to the system.
- **Step-Down Converter:** Reduces the input voltage to a regulated 5V supply for the ESP32 DevKit.

Thus, the project focuses on integrating **IoT-based remote control with mechanical valve actuation** to reduce manual effort and improve convenience in residential waterline operation. The engineering challenge combines embedded systems, electrical motor control, mechanical actuation, and mobile-based communication to address a practical user problem.

2.4 Feasibility Study

The feasibility of the proposed **Automated ON/OFF Switch for Waterline Control** is evaluated through technical, economic, operational, mechanical, safety, and entrepreneurial aspects.

1. Technical Feasibility

The system can be developed using readily available components such as the **ESP32 DevKit, L298N motor driver, 12V DC geared motor, plastic ball valve, step-down converter, and 12V adapter**. The ESP32 provides Wi-Fi communication, while the motor driver controls motor rotation to operate the valve mechanism.

2. Economic Feasibility

The project uses commonly available electronic and mechanical components, making it suitable for a student prototype and a potential low-cost residential automation solution. The overall cost depends on component prices, mechanical mounting, wiring, enclosure, and installation requirements.

3. Operational Feasibility

The proposed system is designed to be operated through a **mobile application**. Users can send ON/OFF commands, which are received by the ESP32 through Wi-Fi. The ESP32 then controls the L298N motor driver to operate the geared motor and change the plastic ball valve position.

4. Mechanical Feasibility

The **12V DC geared motor rated at 30 RPM** provides rotational movement for the valve mechanism. A suitable mechanical coupling or mounting arrangement is required to transfer the motor's rotation to the plastic ball valve. The motor torque and valve operating angle must be verified during prototype testing.

5. Safety Feasibility

The electronic components should be installed in a suitable protective enclosure and kept away from water leakage and moisture. Proper voltage regulation, insulated wiring, secure connections, and appropriate power protection should be provided. The 12V motor supply and 5V ESP32 supply must be connected according to their electrical requirements.

6. Entrepreneurial and Market Feasibility

The project addresses a practical problem experienced by homeowners, tenants, building caretakers, and small residential property managers. A user-friendly, affordable, and easy-to-install product could have potential in the home automation market. Before commercial implementation, the system would require testing for reliability, durability, installation cost, maintenance, and compatibility with different waterline arrangements.

7. Future Scalability

The system can be extended with **water-level sensors, flow sensors, leakage detection, mobile alerts, and automatic control features**. These additions could improve monitoring and support the development of a more comprehensive smart water-management product.

2.5 Methodology Used

The project follows a Design Thinking and MDMD Project-oriented development approach to identify a real user problem, develop a suitable solution, and evaluate its practical potential.

1. **Empathize:** Understand users' difficulties with manual waterline operation through interviews and observation, where conducted.

2. **Define:** Formulate the problem statement based on user needs, focusing on remote control, convenience, affordability, and reduced manual intervention.

3. **Ideate:** Explore alternative solutions, including manual operation, timer-based control, sensor-based automation, and IoT-based mobile control. Select the concept suitable for the project's requirements.

4. **Prototype:** Develop the hardware prototype using the ESP32 DevKit, L298N motor driver, 12V DC geared motor (30 RPM), plastic ball valve, step-down converter, and 12V DC adapter.

5. **Integration:** Connect the hardware and mobile application to enable Wi-Fi-based communication and motorized valve operation.

6. **Test:** Evaluate mobile ON/OFF commands, Wi-Fi communication, motor direction, valve movement, and the stability of the power supply. Identify limitations and areas for improvement.

7. **Entrepreneurial Evaluation:** Consider target users, affordability, ease of installation, maintenance, product reliability, and potential applications in residential buildings.

2.6 Summary

This chapter discussed the problem definition, feasibility study, and Design Thinking and MDMD Project methodology for the proposed Automated ON/OFF Switch for Waterline Control.

The problem definition established the need for a convenient, low-cost, IoT-based system that can remotely control a domestic waterline. The proposed solution integrates an **ESP32 DevKit, Wi-Fi communication, L298N motor driver, 12V DC geared motor rated at 30 RPM, plastic ball valve, step-down converter, and 12V DC adapter**.

The feasibility study considered the technical requirements, cost, operation, mechanical arrangement, electrical safety, entrepreneurial potential, and future scalability of the system. The methodology follows **Empathize, Define, Ideate, Prototype, Integration, and Test**, supported by an entrepreneurial evaluation of user needs and practical product opportunities.

Thus, this chapter establishes the foundation for developing and evaluating the proposed waterline control prototype as a potential home automation solution. The next chapter focuses on **Analysis and Planning**, including project planning, component requirements, task allocation, and project scheduling.

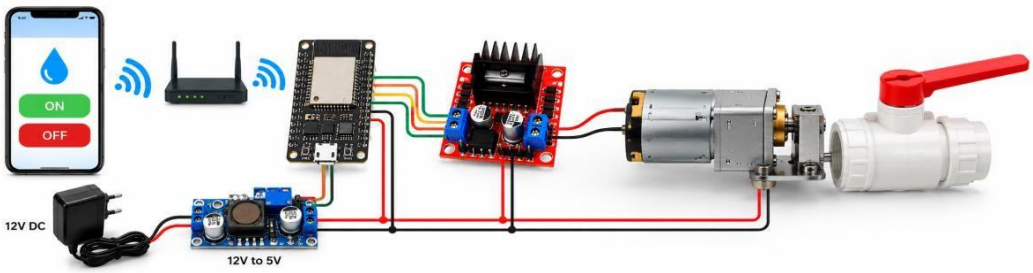


figure 2

CHAPTER 3: ANALYSIS AND PLANNING

3.1 Introduction

This chapter presents the analysis and planning of the proposed Automated ON/OFF Switch for Waterline Control. It identifies user requirements, hardware and software requirements, system components, and the overall project development plan using the Design Thinking and MDMD Project approach.

The system is planned around a mobile application, Wi-Fi communication, an ESP32 DevKit module, an L298N motor driver, a 12V DC geared motor rated at 30 RPM, and a plastic PVC ball valve. A 12V DC adapter supplies power to the system, while a step-down converter provides a regulated 5V supply for the ESP32 DevKit. Proper planning, component selection, prototyping, and testing are required to develop a functional and user-friendly waterline control system.

3.2 USER REQUIREMENT ANALYSIS

The user requirements are identified by considering the difficulties associated with manual waterline operation and the need for convenient remote control.

1. **Remote ON/OFF Control:** Operate the waterline through a mobile application.
2. **Easy Operation:** Provide a simple and user-friendly mobile interface.
3. **Reliable Communication:** Transmit control commands from the mobile application to the ESP32 through Wi-Fi.
4. **Motorized Valve Operation:** Rotate the geared motor according to the received command to operate the plastic ball valve.
5. **Reduced Manual Effort:** Minimize the need for physical operation of the waterline.
6. **Safe Operation:** Protect electronic components from moisture and provide suitable electrical connections and power regulation.
7. **Low-Cost Design:** Use readily available and economical components.
8. **Mechanical Compatibility:** Ensure that the motor and coupling arrangement can operate the plastic ball valve.
9. **Future Expandability:** Allow possible integration of water-level sensors, flow sensors, and smart monitoring features.
10. **User Satisfaction:** Develop a practical solution based on user needs, convenience, affordability, and ease of installation.

3.3 HARDWARE REQUIREMENTS

The major hardware components required for the proposed Automated ON/OFF Switch for Waterline Control are listed below.

Sr. No.	Component	Purpose
1	ESP32 DevKit Module	Main controller for processing commands and Wi-Fi communication
2	L298N Motor Driver	Controls the direction of DC motor rotation
3	12V DC Geared Motor (30 RPM)	Provides mechanical rotation for valve operation
4	Plastic PVC Ball Valve	Controls the opening and closing of the waterline
5	12V DC Adapter	Supplies power to the system
6	Step-Down Converter (12V to 5V)	Provides regulated 5V power for the ESP32 DevKit
7	Connecting Wires and Terminals	Establishes electrical connections between components
8	Mechanical Coupling and Mounting Arrangement	Transfers motor rotation to the valve handle
9	Protective Enclosure	Protects the electronic circuit from moisture and accidental contact
10	Water Pipe / Waterline Setup	Provides the arrangement for demonstrating and testing water flow

table 2

The hardware integrates the **ESP32 DevKit, L298N motor driver, geared motor, and plastic ball valve** to convert a mobile ON/OFF command into physical valve movement. The motor mounting and coupling arrangement must be designed according to the dimensions and torque requirements of the selected valve.

3.4 SOFTWARE AND IoT REQUIREMENTS

The software and IoT requirements support communication between the mobile application and the hardware control system.

Sr. No.	Requirement	Purpose
1	Mobile Application	Provides ON/OFF commands to the user
2	ESP32 Programming Environment	Used to write, upload, and debug the controller program
3	Wi-Fi Network	Enables wireless communication between the application and ESP32
4	ESP32 Firmware	Processes received commands and controls the motor driver
5	IoT Communication Method	Transfers ON/OFF commands between the application and controller
6	Motor Control Logic	Determines the direction and duration of motor operation
7	Mobile User Interface	Provides simple controls for opening and closing the waterline
8	Testing and Debugging Tools	Helps verify command reception, motor operation, and system response

table 3

The software architecture follows the sequence **Mobile Application → Wi-Fi → ESP32 DevKit → L298N Motor Driver → Geared Motor → Plastic Ball Valve**. The exact communication protocol and mobile application platform depend on the software implementation used in the prototype.

3.5 DESIGN THINKING AND PROJECT PLANNING

The project follows the **Design Thinking methodology**, supported by an entrepreneurship-oriented approach to understand user needs, develop a practical prototype, and evaluate its potential for real-world use.

1. Empathize
Understand the difficulties faced by homeowners and other users during manual waterline operation. Interviews and observations, where conducted, help identify user expectations regarding remote control, convenience, affordability, and ease of operation.

2. Define
Analyse the identified user problems and formulate the main problem statement: the need for a convenient, affordable, and remotely controllable domestic waterline system.

3. Ideate
Explore alternative solutions, such as manual control, timer-based automation, sensor-based control, and IoT-based mobile operation. Compare these approaches based on cost, complexity, convenience, and feasibility.

4. Prototype
Develop the selected concept using the ESP32 DevKit, L298N motor driver, 12V DC geared motor rated at 30 RPM, plastic PVC ball valve, step-down converter, and 12V DC adapter. Prepare a suitable mechanical coupling to transfer the motor's rotation to the valve.

5. Test
Evaluate the prototype for Wi-Fi communication, mobile ON/OFF commands, motor direction, valve movement, and power-supply stability. Record issues and identify improvements based on the observed results.

6. Entrepreneurial Evaluation
Identify potential users, including homeowners, tenants, building caretakers, and residential property managers. Evaluate affordability, installation requirements, maintenance, reliability, and possible opportunities for developing the prototype into a practical home automation product.

3.6 PROJECT PLANNING

The project activities are divided into systematic stages to support the development and evaluation of the Automated ON/OFF Switch for Waterline Control.

1. **Problem Identification:** Understand the difficulties associated with manual waterline operation.

2. **User Requirement Analysis:** Identify the needs of users through suitable interviews, observations, and requirement analysis.

3. **Design Thinking:** Apply Empathize, Define, and Ideate to identify the problem and select an appropriate solution.

4. **System Design:** Prepare the system architecture, motor-control logic, electrical connections, and mechanical coupling arrangement.

5. **Component Selection:** Select the ESP32 DevKit, L298N motor driver, 12V geared motor, plastic ball valve, step-down converter, and 12V adapter.

6. **Prototype Development:** Assemble the hardware and prepare the mobile application interface.

7. **System Integration:** Connect the controller, motor driver, power supply, geared motor, and valve mechanism.

8. **Testing and Validation:** Check remote command operation, Wi-Fi communication, motor rotation, and valve movement.

9. **Entrepreneurial Evaluation:** Consider cost, ease of installation, maintenance, user acceptance, and possible future applications.

10. **Documentation:** Record the design, implementation, observations, limitations, and final project results.

3.7 PROJECT SCHEDULING

The following schedule provides a sequence of planned project activities. The actual duration of each activity should be filled in according to the academic project timeline.

Sr. No.	Project Activity	Planned Work
1	Empathize and Problem Identification	Understand user difficulties and requirements
2	Define and Requirement Analysis	Finalize the problem statement and system requirements
3	Literature Survey	Study relevant IoT, motor-control, and waterline automation concepts
4	Ideate and System Design	Compare solutions and prepare the system architecture
5	Component Selection	Finalize hardware and software requirements
6	Prototype Development	Assemble the ESP32, motor driver, motor, and valve mechanism
7	Software and IoT Integration	Implement mobile control and Wi-Fi communication
8	Testing and Validation	Test commands, motor rotation, valve movement, and power supply
9	Result Analysis	Evaluate observations and identify limitations
10	Entrepreneurial Evaluation	Consider affordability, installation, maintenance, and target users
11	Documentation	Prepare the final report and project presentation

table 4

3.8 SUMMARY

This chapter presented the **analysis and planning** of the proposed Automated ON/OFF Switch for Waterline Control. It identified the user requirements, hardware components, software and IoT requirements, and project development activities.

The proposed system uses an **ESP32 DevKit module, L298N motor driver, 12V DC geared motor rated at 30 RPM, plastic PVC ball valve, step-down converter, and 12V DC adapter**. The project planning follows the Design Thinking stages of **Empathize, Define, Ideate, Prototype, and Test**, together with an entrepreneurial evaluation of cost, usability, installation, and potential market applications.

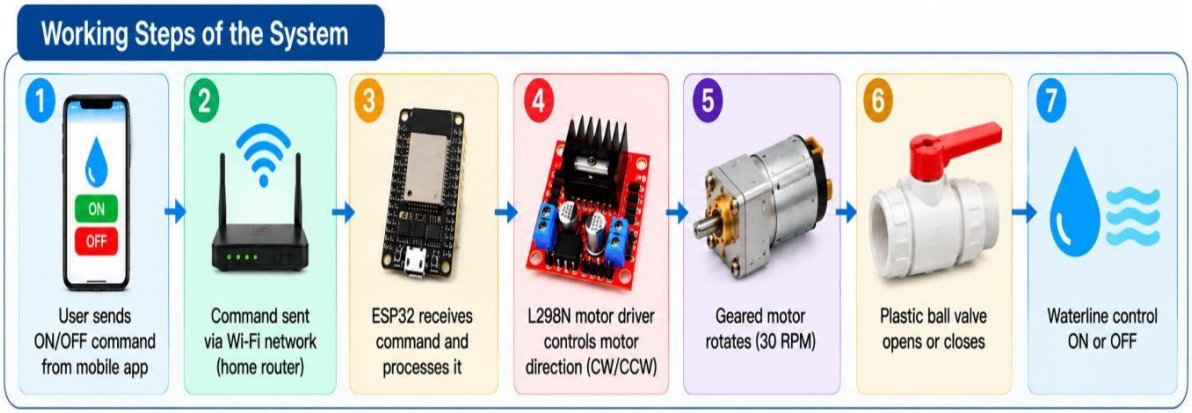


figure 3

CHAPTER 4 – DESIGN & IMPLEMENTATION – PROTOTYPE

4.1 INTRODUCTION

This chapter presents the **design and implementation of the Automated ON/OFF Switch for Waterline Control**. It explains the system architecture, Data Flow Diagram (DFD), flowchart, UML, and prototype development.

The prototype integrates a **Mobile Application, Wi-Fi, ESP32 DevKit module, L298N motor driver, 12V DC geared motor rated at 30 RPM, and plastic PVC ball valve** to achieve remote waterline ON/OFF control. The ESP32 receives commands from the mobile application through Wi-Fi and sends control signals to the L298N motor driver. The motor driver controls the motor's direction of rotation, allowing the motor to operate the connected ball valve.

A **12V DC adapter** supplies power to the system, while a **step-down converter** provides a regulated 5V supply for the ESP32 DevKit. The chapter also explains the hardware assembly and IoT integration used to convert the proposed Design Thinking concept into a practical prototype.

4.2 SYSTEM ARCHITECTURE

The proposed system consists of a **mobile application, Wi-Fi network, ESP32 DevKit, L298N motor driver, geared motor, plastic PVC ball valve, and power supply arrangement**. These components work together to provide remote ON/OFF control of the waterline.

System Working

Mobile Application
↓
Wi-Fi Network
↓
ESP32 DevKit Controller
↓
L298N Motor Driver
↓
12V DC Geared Motor (30 RPM)
↓
Plastic PVC Ball Valve
↓
Waterline ON/OFF

The user sends an ON/OFF command through the mobile application. The command is transmitted through Wi-Fi to the ESP32 DevKit. The ESP32 processes the command and sends control signals to the L298N motor driver. The driver controls the direction of motor rotation, and the geared motor operates the plastic ball valve through a suitable mechanical coupling.

The power supply arrangement uses a **12V DC adapter** for the motor supply and a step-down converter to obtain a regulated 5V supply for the ESP32 DevKit. The final electrical connections must follow the specifications of the selected components.

This architecture integrates **IoT communication, embedded control, motor driving, and mechanical actuation** to provide a practical method for remotely controlling a domestic waterline.

4.3 DFD (DATA FLOW DIAGRAM)

The **Data Flow Diagram (DFD)** represents the flow of commands from the user to the controller and the resulting operation of the waterline control mechanism.

Level 0 DFD

User
↓
ON/OFF Command
↓
Mobile Application
↓
Wi-Fi Network
↓
ESP32 DevKit Controller
↓
L298N Motor Driver
↓
Geared Motor
↓
Plastic PVC Ball Valve
↓
Waterline ON/OFF

The user sends an ON/OFF command through the mobile application. The command is transmitted through Wi-Fi to the ESP32. The controller processes the command and sends the appropriate signals to the L298N motor driver. The motor driver operates the geared motor, which rotates the connected valve mechanism to control water flow.

Data Flow

- **Input:** ON/OFF command from the user.
- **Communication:** Wi-Fi network.
- **Processing:** ESP32 DevKit controller.
- **Motor Control:** L298N motor driver.
- **Actuation:** 12V DC geared motor rated at 30 RPM.
- **Mechanical Control:** Plastic PVC ball valve.
- **Output:** Waterline opened or closed.
-

The DFD explains how the user's command is converted into physical valve movement through electronic processing and mechanical actuation.

4.4 BLOCK DIAGRAM

The block diagram represents the main functional units and their relationships in the proposed system.

Input Unit: Mobile Application
↓
Communication Unit: Wi-Fi Network
↓
Controller Unit: ESP32 DevKit
↓
Motor Driver Unit: L298N Motor Driver
↓
Actuator Unit: 12V DC Geared Motor (30 RPM)
↓
Output Unit: Plastic PVC Ball Valve and Waterline
Power Supply Unit: 12V DC Adapter → Motor Supply and Step-Down Converter → Regulated 5V Supply for ESP32 DevKit.

The controller processes the received command, while the motor driver controls the direction of motor rotation. The mechanical arrangement transfers the motor's movement to the plastic ball valve to open or close the waterline.

4.5 FLOWCHART

The flowchart represents the step-by-step operation of the Automated ON/OFF Switch for Waterline Control.

System Flow

START
↓
Power ON the System
↓
Initialize ESP32 DevKit
↓
Connect to Wi-Fi Network
↓
Open Mobile Application
↓
Select ON/OFF Command
↓
Command Sent to ESP32
↓
ESP32 Receives and Processes Command
↓
Send Control Signal to L298N Motor Driver
↓
Motor Rotates in the Required Direction
↓
Plastic Ball Valve Opens or Closes
↓
Waterline ON/OFF
↓
END / Wait for Next Command

The system begins by powering the controller and establishing Wi-Fi communication. Once a command is received, the ESP32 sends the required motor-control signal to the L298N driver. The geared motor then operates the valve mechanism.

The motor's direction and operating duration must be configured and tested according to the actual mechanical arrangement. If valve-position feedback is not installed, the system should not claim to confirm the valve's actual position automatically.

4.6 UML

The **Unified Modeling Language (UML)** describes the interactions between the user, mobile application, ESP32 controller, motor driver, motor, and valve mechanism.

Use Case Diagram

Actor: User
User Actions:

- Open Mobile Application
- Connect to the IoT System
- Send ON Command
- Send OFF Command
- Operate the Waterline Remotely

System Functions:

- Establish Wi-Fi Communication
- Receive Command through ESP32
- Process the Control Command
- Send Signals to the L298N Motor Driver
- Operate the Geared Motor
- Open or Close the Plastic PVC Ball Valve

Sequence Flow

User → Mobile Application → Wi-Fi → ESP32 DevKit → L298N Motor Driver → Geared Motor → Plastic PVC Ball Valve → Waterline

The user sends an ON/OFF command through the mobile application. The ESP32 receives and processes the command, sends control signals to the motor driver, and the geared motor operates the valve mechanism. This sequence represents the proposed interaction between the software, electronic controller, and mechanical output.

4.7 PROTOTYPE DEVELOPMENT

The prototype of the **Automated ON/OFF Switch for Waterline Control** was developed by integrating the IoT control system with a motorized mechanical valve arrangement. The main components used are the **ESP32 DevKit, L298N motor driver, 12V DC geared motor, plastic PVC ball valve, step-down converter, 12V DC adapter, connecting wires, and mobile application**.

The prototype development is divided into the following stages.

4.7.1 ESP32 and Wi-Fi Integration

The ESP32 DevKit is programmed as the main IoT controller. It uses its built-in Wi-Fi capability to communicate with the selected mobile application or control interface. The firmware receives and processes ON/OFF commands and generates the required motor-control signals.

4.7.2 L298N Motor Driver Integration

The L298N motor driver is connected to the ESP32 and the DC geared motor according to the circuit design. It controls the direction of motor rotation based on the input signals from the ESP32. The driver is used instead of a relay module because the proposed prototype requires directional control of the DC motor.

4.7.3 Motor and Plastic Ball Valve Mechanism

A **12V DC geared motor rated at 30 RPM** provides the rotational movement required to operate the plastic PVC ball valve. A suitable coupling or mounting arrangement transfers the motor's rotation to the valve handle.
The motor torque, valve operating angle, and mechanical alignment must be verified during testing to ensure that the valve can reach its open and closed positions without damaging the mechanism.

4.7.4 Power Supply Integration

A **12V DC adapter** provides the required power input for the system. A step-down converter reduces the voltage to a regulated 5V supply for the ESP32 DevKit. The motor driver and motor must be connected according to their rated voltage and current requirements.

Suitable grounding, insulated connections, and protection from water and moisture are required for reliable operation.

4.7.5 Mobile Application and IoT Control

The mobile application provides ON and OFF controls for the user. Commands are transmitted wirelessly to the ESP32 through the selected Wi-Fi-based communication method. The controller then sends the corresponding signals to the L298N motor driver to operate the geared motor.

4.7.6 Hardware Assembly

All the components are assembled and connected to form the complete prototype. The final arrangement integrates **mobile control, Wi-Fi communication, embedded processing, motor driving, power conversion, and mechanical valve actuation** into a single system.

The assembled prototype is evaluated to check communication, motor rotation, valve movement, and the overall stability of the control arrangement.

4.8 GEOTAGGED PHOTOS AND USER INTERVIEWS

This section presents the field study and user interaction associated with the development of the Automated ON/OFF Switch for Waterline Control.

4.8.1 User Interview

User interviews, where conducted, help identify the problems associated with manual waterline operation, including the need for physical presence, inconvenience, difficulty in remote operation, and unnecessary water flow.

Interview Focus:

- Problems with manual valve operation.
- Need for remote ON/OFF control.
- User convenience and safety.
- Acceptance of mobile-based control.
- Affordability and ease of installation.
- Suggestions for future improvements.
-

The information collected during the Empathize stage of Design Thinking supports the definition of user requirements and the selection of the proposed IoT-based solution.

4.8.2 Geotagged Photos

Geotagged photographs of the existing waterline setup, component assembly, and prototype development can be included to document the field study and practical implementation.

The photographs should represent the actual location and prototype. Captions may identify the existing waterline arrangement, electronic component assembly, motor-to-valve coupling, and final prototype testing. Only photographs actually captured during the project should be presented as project evidence.

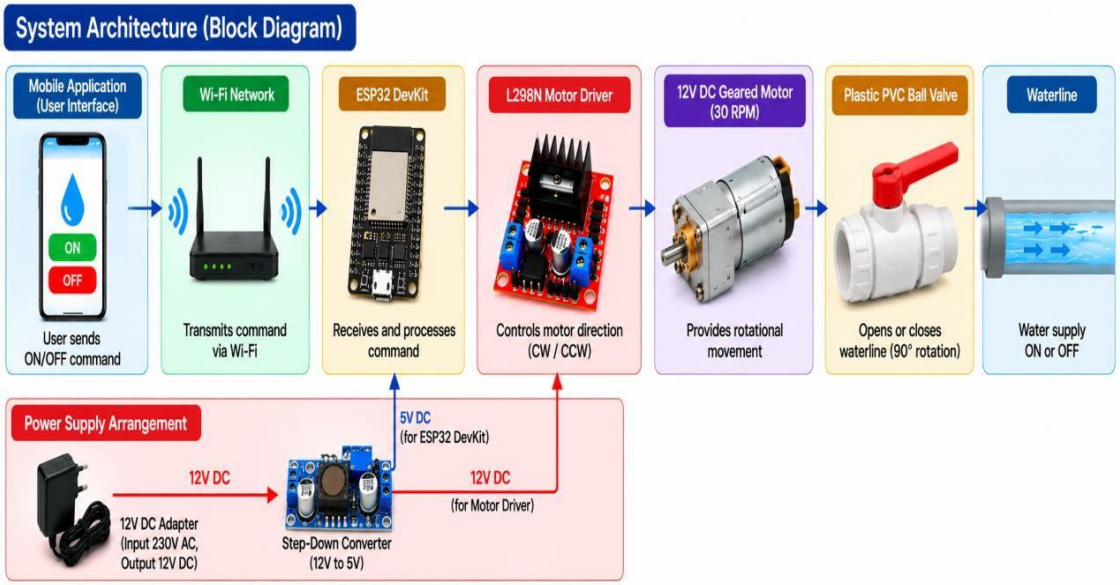


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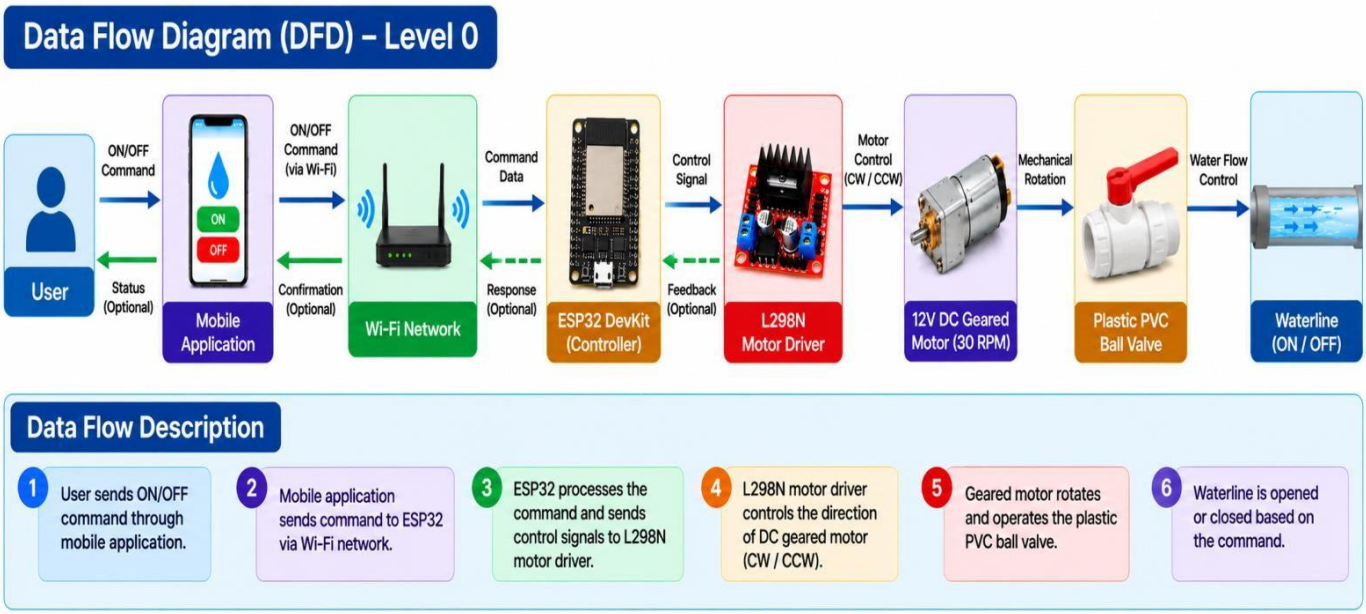


figure 4.2

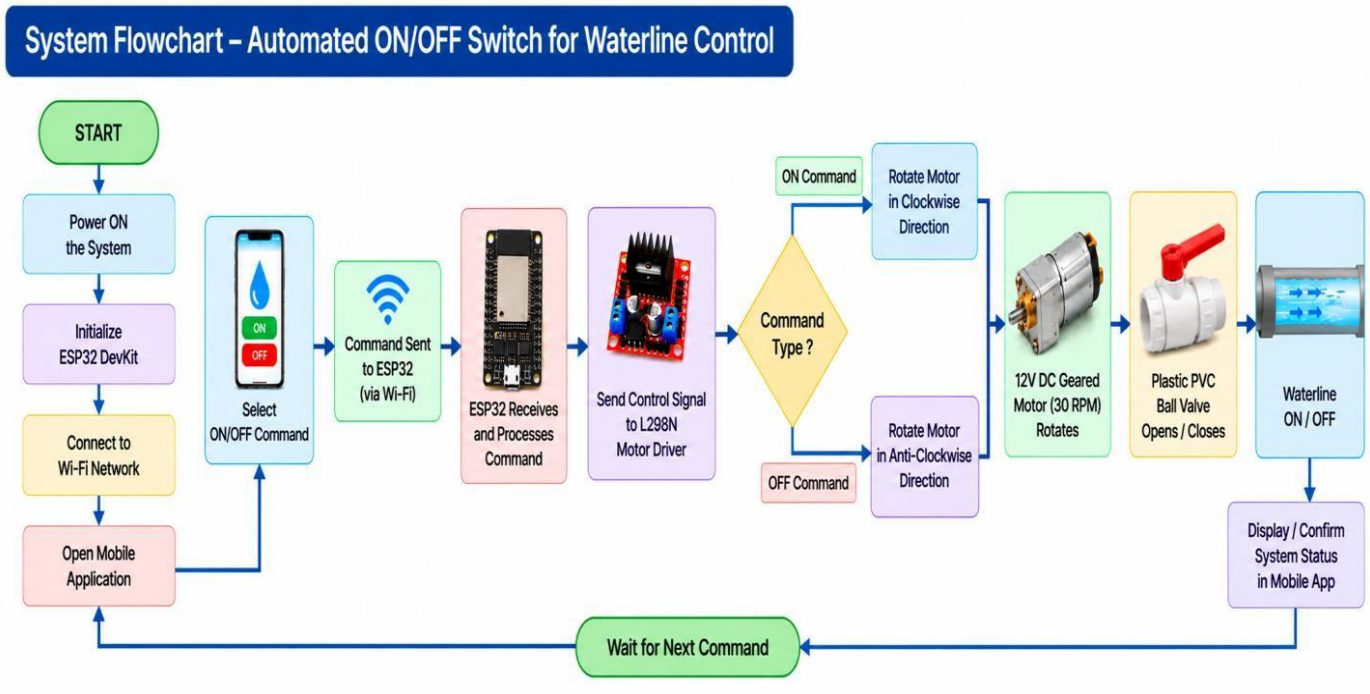


figure 4.3

CHAPTER 5 – TESTING, RESULTS & DISCUSSION

5.1 TEST STAGE – PROTOTYPE TESTING

The prototype of the **Automated ON/OFF Switch for Waterline Control** is evaluated to verify IoT communication, motor control, and valve operation. Testing focuses on mobile commands, ESP32 communication, L298N motor driver response, motor rotation, and opening/closing of the plastic PVC ball valve.

The system uses an **ESP32 DevKit, L298N motor driver, 12V DC geared motor (30 RPM), plastic PVC ball valve, 12V DC adapter, and step-down converter providing 5V for the ESP32 DevKit**. Testing follows the Design Thinking Test stage to evaluate the prototype against user requirements.

5.1.1 REMOTE ON/OFF TESTING

- **ON Command:** The mobile application sends a command through Wi-Fi to the ESP32, which controls the L298N to rotate the motor towards the valve-open position.
- **OFF Command:** The ESP32 controls the motor driver to move the valve towards the closed position
- **Expected Result:** The valve should reach the intended position without manual operation.

This test evaluates the main requirement of remote waterline control.

5.1.2 ESP32–Wi-Fi COMMUNICATION TESTING

- **Connection:** Check whether the ESP32 connects to the configured Wi-Fi network.
- **Command Transmission:** Send ON/OFF commands through the mobile application.
- **Command Reception:** Verify that the ESP32 processes the commands correctly.
- **Expected Result:** Commands should be received when the network and application are available.

5.1.3 L298N MOTOR DRIVER TESTING

The L298N motor driver is evaluated to verify directional control of the 12V DC geared motor

- **Forward Rotation:** Check motor movement towards valve opening.
- **Reverse Rotation:** Check motor movement towards valve closing.
- **Expected Result:** The motor should rotate in the required direction within the components' operating limits.

The L298N provides directional motor control rather than relay-based switching.

5.1.4 MOTOR AND PLASTIC BALL VALVE TESTING

The mechanical arrangement is evaluated to verify operation of the plastic PVC ball valve.

- Check whether the 30 RPM geared motor moves the valve in both directions.
- Inspect motor-to-valve coupling and mechanical alignment.
- Verify that the motor provides sufficient torque.

Expected Result: The valve should move smoothly towards the intended open or closed position.

Without position sensors or limit switches, the system cannot independently confirm the exact valve position.

5.1.5 POWER SUPPLY TESTING

- **12V DC Adapter:** Verify the required input voltage and current capacity.
- **Step-Down Converter:** Check the regulated 5V supply for the ESP32 DevKit.
- **Motor Supply:** Verify connections according to the motor and driver ratings.
- **Expected Result:** The supply should remain stable without unexpected resets or overheating.

All electrical connections must be protected from moisture and follow component specifications.

5.1.6 END-TO-END SYSTEM TESTING

Mobile Application → Wi-Fi → ESP32 DevKit → L298N Motor Driver → 12V Geared Motor → Plastic PVC Ball Valve → Waterline

The user sends an ON/OFF command through the mobile application. The ESP32 processes it, the L298N controls motor direction, and the motor operates the valve.

Expected Result: The system should execute the command and move the valve towards the intended position.

5.2 RESULTS AND DISCUSSION

The prototype is intended to demonstrate remote waterline control through IoT communication and mechanical actuation. The following parameters are considered during evaluation.

Sr. No.	Test Parameter	Evaluation Criteria
1	Wi-Fi Connectivity	ESP32 connects to the network
2	Command Reception	ON/OFF command is processed
3	Motor Driver	L298N controls motor direction
4	Motor Rotation	Motor rotates as required
5	Valve Operation	Valve moves towards the intended position
6	Power Stability	No unexpected power interruptions
7	Remote Operation	Commands can be sent without manual valve operation

table 5

Actual observations should be recorded after testing.

Sr. No.	Test	Actual Observation	Status
1	Wi-Fi connection	working	Pass
2	ON/OFF commands	working	Pass
3	Motor direction	working	Pass
4	Valve movement	working	Pass
5	Power stability	working	Pass

table 6

The results help identify communication issues, mechanical limitations, and power-related problems. In the Design Thinking Test stage, these observations and user feedback guide improvements to the prototype.

5.3 ADVANTAGES

1. **Remote Operation:** Mobile-based waterline control.
2. **Reduced Manual Effort:** Less need for physical valve operation.
3. **IoT-Based Control:** ESP32 provides Wi-Fi communication.
4. **Directional Motor Control:** L298N controls motor rotation.
5. **Motorized Valve:** Enables operation of a plastic PVC ball valve.
6. **Practical Design:** Integrates electronic and mechanical components.
7. **Potential Water Savings:** Timely closure may reduce unnecessary flow.
8. **Affordable Prototype:** Uses commonly available components.
9. **Expandability:** Supports future sensor integration.
10. **Entrepreneurial Potential:** May be developed into a residential automation product.

5.4 LIMITATIONS

1. **Wi-Fi Dependency:** Remote operation requires network availability.
2. **Power Dependency:** The controller and motor require a suitable supply.
3. **Limited Automation:** The current design mainly provides remote ON/OFF control.
4. **Mechanical Alignment:** Proper motor-to-valve coupling is essential.
5. **Motor Torque:** The motor must provide sufficient torque.
6. **Valve Compatibility:** Different valve sizes may require mounting changes.
7. **Limited Monitoring:** Water-level and flow sensors are not included.
8. **Position Confirmation:** Sensors are required for independent valve-position feedback.
9. **Electrical Safety:** Proper insulation and enclosure are necessary.
10. **Commercial Readiness:** Durability and reliability require further testing.

5.5 FUTURE SCOPE

The system can be enhanced through the following features:

1. **Water-Level Sensors:** Monitor tank water levels.
2. **Flow Sensors:** Measure water flow.

3. **Automatic Control:** Operate the valve according to sensor readings.
4. **Mobile Notifications:** Provide system alerts.
5. **Position Feedback:** Add limit switches or position sensors.
6. **Smart Scheduling:** Enable scheduled operation.
7. **IoT Dashboard:** Display status and sensor data.
8. **Fault Detection:** Identify motor or communication failures with suitable sensing.
9. **Multi-Valve Control:** Operate multiple waterlines.
10. **Cloud Integration:** Store and analyse water-usage data.

These improvements could develop the prototype into a more comprehensive smart water-management system.

5.6 ENTREPRENEURSHIP AND MARKET POTENTIAL

The project addresses the inconvenience of manual waterline operation and may be useful for homeowners, tenants, caretakers, and residential property managers.

Target Customers

- Homeowners seeking remote valve control.
- Residential buildings with inconvenient valve access.
- Building caretakers and small property managers.

Value Proposition

The system aims to provide **remote control, reduced manual effort, and motorized valve operation** using readily available components.

Cost and Affordability

The prototype cost depends on the ESP32 DevKit, L298N, motor, plastic PVC ball valve, power supply, wiring, enclosure, mounting, and installation.

Commercial Challenges

Commercial deployment requires reliability testing, valve compatibility checks, electrical safety, moisture protection, maintenance planning, and customer feedback. The Design Thinking process supports iterative improvements based on user requirements and actual test observations.

5.7 DEPLOYMENT

Deployment may be considered after successful prototype testing.

1. **Site Assessment:** Select a compatible waterline and plastic PVC ball valve.
2. **Mechanical Installation:** Secure the motor and valve coupling.
3. **Controller Installation:** Place the ESP32 and L298N inside a protective enclosure.
4. **Power Installation:** Connect the 12V adapter and 5V step-down supply correctly.
5. **IoT Configuration:** Configure Wi-Fi and the mobile application.
6. **Functional Testing:** Verify commands, motor direction, and valve movement.
7. **Safety Verification:** Check insulation, moisture protection, and mechanical mounting.
8. **User Demonstration:** Explain operation and maintenance.

Potential applications include homes, apartments, and residential buildings requiring remote valve operation. Practical deployment requires suitable safety and reliability checks.

5.8 SUMMARY

This chapter presented the testing approach, result-evaluation criteria, advantages, limitations, future scope, entrepreneurial potential, and deployment of the **Automated ON/OFF Switch for Waterline Control**.

The prototype integrates the **ESP32 DevKit, L298N motor driver, 12V DC geared motor rated at 30 RPM, plastic PVC ball valve, 12V DC adapter, and step-down converter providing 5V for the ESP32**.

The testing process evaluates Wi-Fi communication, command processing, motor direction, valve operation, and power stability. The Design Thinking Test stage helps identify improvements, while the MDMD Project evaluation considers affordability, target customers, installation, and commercial potential.

Actual observations must be recorded before confirming successful performance. Further safety, reliability, and usability testing can help develop the prototype into a practical IoT-based home automation solution.

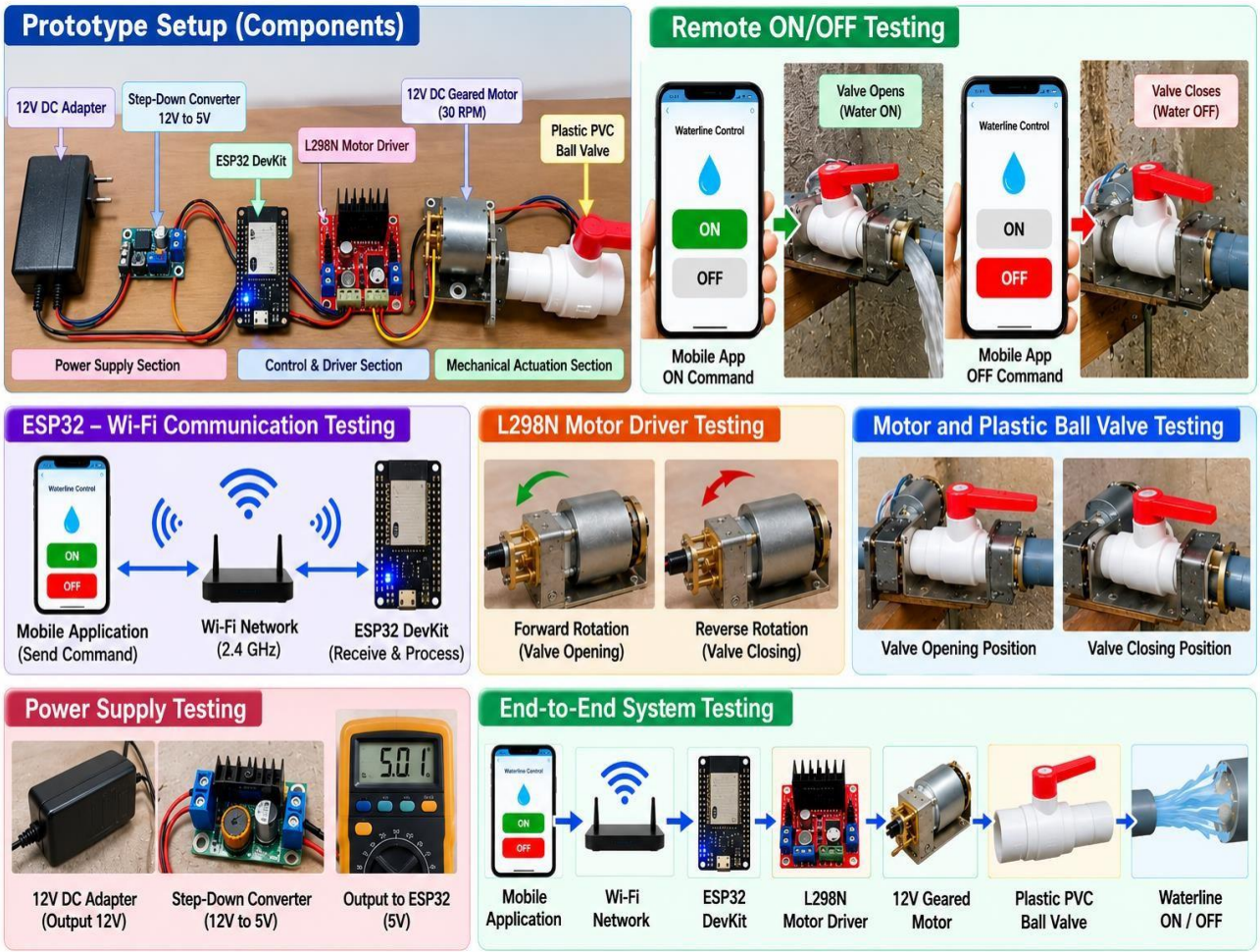


figure 5

CHAPTER 6 – FINAL SOLUTION & CONCLUSION

6.1 FINAL PROPOSED SOLUTION

The final proposed solution is an IoT-based Automated ON/OFF Switch for Waterline Control that enables users to control a domestic waterline remotely through a mobile application.

The system integrates an ESP32 DevKit, Wi-Fi communication, L298N motor driver, 12V DC geared motor rated at 30 RPM, plastic PVC ball valve, 12V DC adapter, and step-down converter. The ESP32 receives ON/OFF commands through Wi-Fi and sends control signals to the L298N motor driver. The driver controls the direction of motor rotation, allowing the motor to operate the connected ball valve.

The 12V DC adapter supplies power to the system, while the step-down converter provides a regulated 5V supply for the ESP32 DevKit, according to the selected hardware configuration.

Final System Flow

Mobile Application → Wi-Fi → ESP32 DevKit → L298N Motor Driver → 12V Geared Motor → Plastic PVC Ball Valve → Waterline ON/OFF
The proposed solution combines IoT communication, embedded systems, electrical motor control, and mechanical actuation to reduce manual effort and improve convenience in residential waterline operation.

6.2 DESIGN THINKING JOURNEY

The project follows the Design Thinking methodology to understand user needs, define the problem, develop a suitable concept, and evaluate the prototype.

1. **Empathize**: Identify the inconvenience of manual waterline operation and the need for remote control.
2. **Define**: Establish the requirement for a convenient, affordable, and user-friendly waterline control system.
3. **Ideate**: Explore manual, timer-based, sensor-based, and IoT-based solutions and select mobile-controlled valve operation.
4. **Prototype**: Integrate the ESP32 DevKit, L298N motor driver, 12V geared motor, plastic PVC ball valve, power supply, and mobile application.
5. **Test**: Evaluate Wi-Fi communication, command reception, motor direction, valve movement, and power stability.
6. **Improve**: Identify mechanical, electrical, and communication limitations and consider improvements based on testing and user feedback.
   This process demonstrates how a practical user problem can be converted into an engineering solution through user-centred design, prototyping, testing, and iterative improvement.

6.3 ENTREPRENEURSHIP AND COMMERCIAL POTENTIAL

The project addresses a practical problem faced by homeowners, tenants, building caretakers, and residential property managers who require convenient access to waterline controls.

Value Proposition: Remote waterline operation through a mobile application using commonly available electronic and mechanical components.
Target Customers: Individual homeowners, apartments, residential societies, and small property managers.

Affordability: The prototype uses readily available components. The final product cost would depend on hardware, mounting, protective enclosure, installation, and maintenance requirements.

Future Business Opportunity: The concept could be developed into a residential waterline automation product with optional water-level monitoring, flow sensing, notifications, and automatic control.

Before commercialisation, the product would require further reliability testing, electrical and moisture protection, compatibility checks, customer feedback, and cost evaluation.

6.4 CONCLUSION

The Automated ON/OFF Switch for Waterline Control presents an IoT-based approach to remotely operating a domestic waterline. The system integrates an ESP32 DevKit, Wi-Fi, L298N motor driver, 12V DC geared motor rated at 30 RPM, plastic PVC ball valve, 12V DC adapter, and step-down converter.

The proposed system converts mobile application commands into motor movement, enabling the connected valve to move towards its open or closed position. It demonstrates the integration of IoT communication, embedded control, and mechanical actuation to reduce manual intervention and improve user convenience.

The Design Thinking process supports the identification of user requirements, selection of a suitable solution, prototype development, and testing. From an entrepreneurial perspective, the project has potential as an affordable residential automation concept, subject to further safety, durability, reliability, and market evaluation.

Future improvements may include water-level sensors, flow monitoring, valve-position feedback, mobile notifications, and automatic control. Thus, the project provides a foundation for a more comprehensive smart water-management system.

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